

Miniaturized Implantable MIMO Antenna for Gastrointestinal Wireless Capsule Endoscopy

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Abstract—This study presents a miniaturized triple-band MIMO implantable antenna designed for wireless capsule endoscopy. The antenna operates in the Industrial, Scientific, and Medical (ISM) bands (2.4–2.4835 GHz and 5.725–5.875 GHz) for data telemetry and the Medical Telemetry Service (WMTS) band (1.395–1.4 GHz) for wireless power transfer. The antenna has a compact footprint of $8.5 \times 4 \times 0.26 \text{ mm}^3$. The design is optimized within a realistic capsule device, incorporating batteries, sensors, and electronic components. Simulation results demonstrate good performance, including reflection coefficients of -29 dB , -15 dB , and -36 dB ; gains of -40 dBi , -28 dBi , and -15 dBi ; and -10 dB impedance bandwidths of 260 MHz, 170 MHz, and 160 MHz at 1.4 GHz, 2.45 GHz, and 5.8 GHz resonating frequencies respectively. Mutual coupling remains below -46.74 dB , -32.24 dB , and -22.67 dB across the bands. SAR compliance confirms the antenna safety for in-body use. Link-budget analysis further verifies reliable high-data-rate 100 Mbps communication up to 20 m at the respective resonant frequencies, highlighting its suitability for advanced high-speed medical telemetry.

I. INTRODUCTION

In recent years, implantable and ingestible medical devices have become increasingly prevalent in modern healthcare, enabling critical applications such as wireless capsule endoscopy (WCE), leadless cardiac pacemaker, intracranial pressure monitoring, and blood glucose monitoring [1–3]. These implantable biomedical devices (IMBDs) serve several purposes: they collect physiological data from various body tissues to facilitate accurate disease diagnosis while simultaneously transmitting this information wirelessly to medical centers. WCE systems, in particular, have transformed gastrointestinal (GI) tract diagnosis by allowing non-invasive detection and assessment of various infections and disorders. However, developing effective antennas for WCE applications presents several technical challenges, including size constraints, rigorous safety requirements, and frequency detuning effects during GI tract traversal, the need for adequate gain in lossy biological tissues, and the demand for high data-rate transmission. Current solutions predominantly employ single-input-single-output (SISO) configurations, which inherently limit spectral efficiency and consequently constrain achievable data rates. This limitation highlights the critical need for more advanced antenna design in next-generation WCE systems to overcome these fundamen-

tal performance limitations. Furthermore, conventional SISO antennas suffer from an inherent inability to mitigate multipath propagation effects, which severely limits channel capacity. This limitation becomes particularly critical with the advent of next-generation CMOS-based cameras, which enable real-time transmission of high-resolution images and video streams requiring data rates of 78 Mbps or higher. The multi-input multi-output (MIMO) approach offers a compelling solution, not only enhancing data rates for high-quality image transmission but also effectively addressing multipath distortion to improve overall communication reliability. As WCE and IMBDs increasingly incorporate multifunctional capabilities such as sleep/wake-up signaling, wireless power transfer, and biotelemetry, single or dual-band antennas are no longer sufficient. These advanced applications demand simultaneous operation across multiple frequency bands, driving the need for innovative multiband MIMO antenna designs that can support the complex requirements of next-generation IMBDs while maintaining optimal performance in challenging biological environments.

To overcome this, in this study, we proposed a miniaturized triple-band implantable MIMO antenna specifically designed for WCE applications. The proposed antenna operates at three frequencies including 1.4 GHz, 2.45 GHz, and 5.8 GHz to support biotelemetry and wireless power transfer. With an ultra-compact volume of just 8.84 mm^3 , the antenna achieves high inter-element isolation while maintaining optimal performance across all critical parameters, including reflection coefficient, gain, and -10 dB impedance bandwidth. Rigorous evaluations of SAR compliance and wireless communication performance demonstrate the design efficacy and safety for biomedical implantation.

II. MIMO ANTENNA DESIGN AND STRUCTURE

The design of implantable antennas presents unique challenges that fundamentally differ from free-space antenna design, necessitating specialized simulation setup. In this study, the proposed antenna was modeled and evaluated using ANSYS High-Frequency Structure Simulator (HFSS). To accurately represent implantation scenario, the antenna was

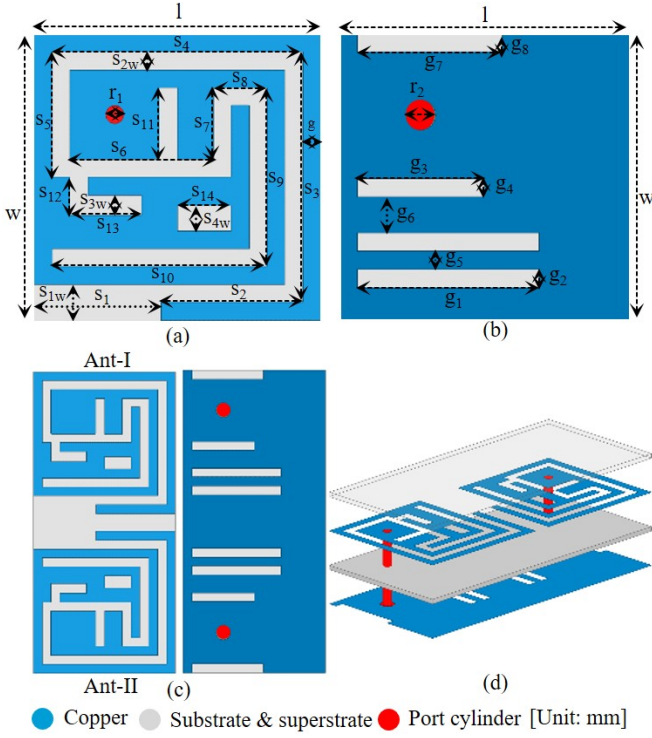


Fig. 1. (a) SISO patch (b) SISO ground (c) top and bottom view of MIMO (d) exploded 3D view of MIMO ($l = w = 4$, $S_1 = 1.77$, $S_{1w} = 0.5$, $S_2 = 1.52$, $S_3 = S_4 = 3.5$, $S_5 = 1.75$, $S_6 = 2.5$, $S_7 = S_{11} = 1.25$, $S_8 = 0.75$, $S_9 = 2.5$, $S_{10} = 3$, $S_{12} = 0.78$, $S_{13} = 1$, $S_{14} = 0.75$, $g = 0.25$, $S_{2w} = S_{3w} = 0.25$, $S_{4w} = 0.35$, $g_1 = g_5 = 2.5$, $g_3 = 1.75$, $g_7 = 2$, $g_2 = g_4 = g_6 = g_8 = 0.25$, $g_6 = 0.5$, $r_1 = 0.125$, $r_2 = 0.2$).

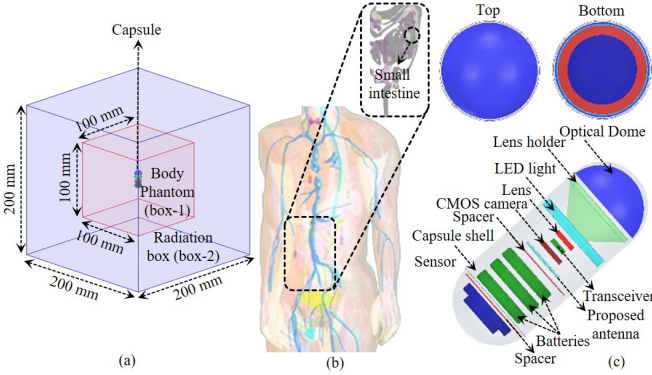


Fig. 2. (a) Homogeneous simulation setup (HFSS) (b) heterogeneous (Sim4life) (c) wireless capsule endoscopy.

positioned at the center of a homogeneous small intestine phantom, selected due to its anatomical complexity within the GI tract system. The phantom designated as Box-1, measures $100 \times 100 \times 100 \text{ mm}^3$ and is enclosed within a radiation boundary box (Box-2) of $200 \times 200 \times 200 \text{ mm}^3$ dimensions, as depicted in Fig. 2 (a). The antenna was simulated at an implantation depth of 50 mm to reflect realistic conditions. The simulation incorporates frequency-dependent dielectric properties of human tissue to enhance biological fidelity: the small intestine phantom exhibits relative permittivity values

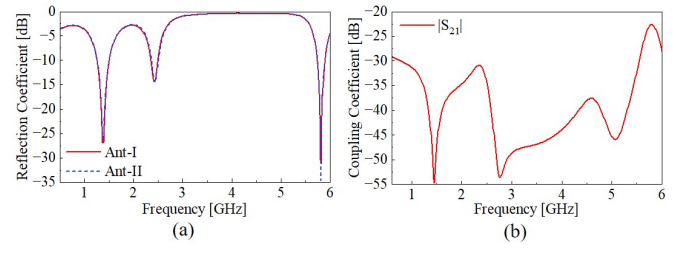


Fig. 3. (a) Reflection coefficient (b) coupling coefficient.

of 57.1, 54.4, and 48.7 at 1.4 GHz, 2.45 GHz, and 5.8 GHz respectively, with corresponding electrical conductivities of 2.44 S/m, 3.17 S/m, and 6.75 S/m at these frequencies. The developed simulation setup enables accurate characterization of antenna performance in biologically relevant conditions. To ensure biocompatibility, the proposed antenna is encapsulated within a WCE device lined with alumina (Al_2O_3), characterized by a dielectric constant of $\epsilon_r = 9.8$ and a loss tangent of $\tan\delta = 0.006$, thereby minimizing potential tissue interactions. The cylindrical capsule enclosure measures 11 mm in diameter with 0.25 mm wall thickness and 24.65 mm length, as illustrated in Fig. 2 (c). Fig. 1 present detailed 2D and 3D exploded views comparing both SISO and MIMO configurations. The MIMO antenna design incorporates four components including meandered radiating patch, slotted ground plane, substrate, and a protective superstrate. Roger RT/duroid 6010, characterized by a dielectric permittivity of $\epsilon_r = 10.2$ and a loss tangent of $\tan\delta = 0.0023$, is employed for substrate and superstrate. To enhance isolation from internal capsule circuitry while maintaining compactness, the design employs a superstrate layer matching the substrate dimensions and material properties. The antenna is excited through a 50Ω coaxial feed port.

III. RESULTS AND DISCUSSION

As demonstrated in Fig. 3 (a), both elements of the MIMO antenna system exhibit consistent resonance characteristics, with reflection coefficients below -29 dB , -15 dB , and -36 dB at 1.4 GHz, 2.45 GHz, and 5.8 GHz, respectively. The coupling coefficients $|S_{21}| = |S_{12}|$ remain low, measuring below -46 dB , -32 dB , and -22 dB across all resonant frequencies, as depicted in Fig. 3 (b). The antenna achieves good -10 dB bandwidths of 260 MHz, 170 MHz, and 160 MHz at the respective frequencies. Radiation pattern analysis, illustrated in Fig. 4, reveals E-plane and H-plane performance at all operational frequencies, with gains of -40 dBi , -28 dBi , and -15 dBi . At lower frequency 1.4 GHz, the antenna gains is comparatively lower than 2.45 GHz and 5.8 GHz. Biological tissues like small intestine exhibit higher dielectric losses due to higher dielectric permittivity values at lower frequency, which leads to greater energy loss within the body rather than efficient radiation. As a consequence, the signal is absorbed more by the surrounding tissues, thereby reducing the antenna gain. To ensure compliance with biomedical safety standards, the SAR was evaluated under realistic implantation conditions. Simulations conducted in a heterogeneous human

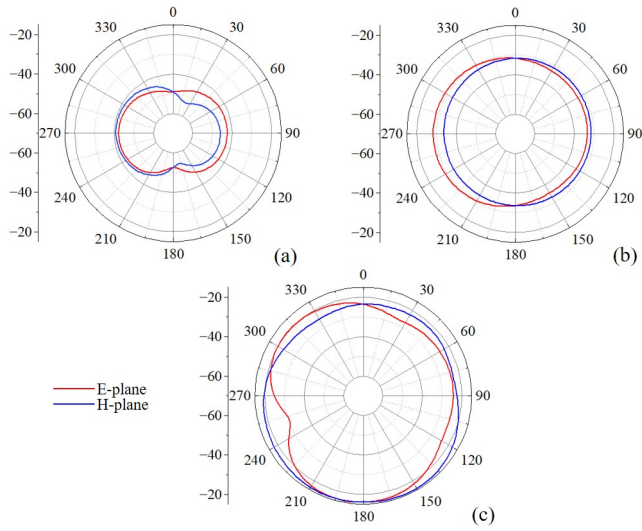


Fig. 4. Radiation pattern of Ant-I (a) 1.4 GHz (b) 2.45 GHz (c) 5.8 GHz.

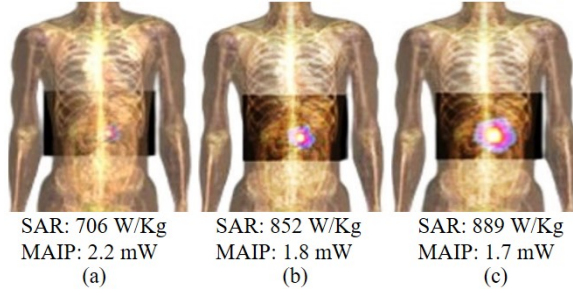


Fig. 5. Normalized 1-g SAR distribution and MAIP of the proposed MIMO antenna (a) 1.4 GHz (b) 2.45 GHz (c) 5.8 GHz

tissue model, with the setup configuration illustrated in Fig. 2 (b), maintaining the same implant depth as the homogeneous setup confirm adherence to IEEE C95.1-2005 guidelines, with peak SAR values below the permissible limits of 1.6 W/kg (1-g tissue) [8]. Each antenna element was excited with 1 W input power, and the resulting SAR values along with maximum allowable input power (MAIP) thresholds are presented in Fig. 5. Importantly, the antenna demonstrates full compliance with biomedical safety standards, operating well within the -16 dBm power limitation for implantable devices across all target frequency bands. For telemetry performance evaluation, we calculated link margin analysis of proposed MIMO antenna. As illustrated in Fig. 6 (b), the proposed MIMO antenna maintains robust communication links supporting data rates up to 100 Mbps over distances exceeding 20 meters at all operational frequencies.

IV. CONCLUSION AND FUTURE WORK

A miniaturized triple-band implantable MIMO antenna has been developed for WCE applications, operating at 1.4 GHz, 2.45 GHz, and 5.8 GHz. The compact design incorporates two antenna elements with total dimensions of $8.5 \times 4 \times 0.26$ mm³, resulting in a total volume of 8.84 mm³. Simulation results demonstrate good performance across key parameters,

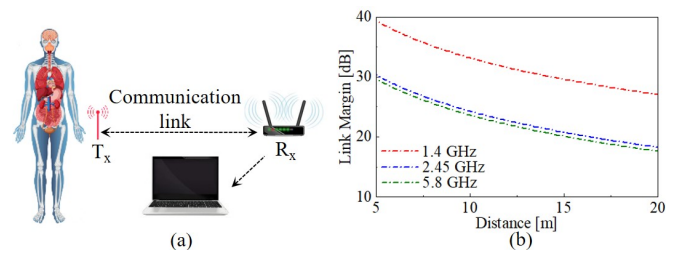


Fig. 6. Link margin analysis at 100 Mbps.

TABLE I
COMPARISON WITH RECENTLY REPORTED IMPLANTABLE ANTENNAS

Ref.	Size [mm ³]	Freq [GHz]	Gain [dBi]	Iso. [dB]	Depth [mm]	No. of elements
[4]	14.69	0.915	-23.23	21.77	60	2
[5]	13.03	0.433	-30	26	5	2
[6]	15.36	0.95, 2.45	-32.15, -22.2	24	58	2
[7]	444	2.45	-20	26.3	50	2
This work	8.84	1.4, 2.45, 5.8	-40, -28, -15	48.9, 32.2, 22.6	50	2

including reflection coefficient, isolation, gain, and bandwidth. SAR analysis confirms the antenna compliance with biomedical safety standards. Future work will focus on fabrication, experimental validation, and investigation of MIMO channel characteristics.

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REFERENCES

- [1] T. Bashir, W. Li and T. Xia, "Radiation Diversity Enabled Self-Isolated Compact Dual-Band Cubic MIMO Antenna for Wireless Biomedical Implants in Variable and Dynamic Environment," *IEEE J. Microwaves*, doi: 10.1109/JMW.2025.3583891.
- [2] N. Abbas, S. Ullah, Z. Bashir, A. Basir and H. Yoo, "Design and Measurement of a Minuscule-Sized Implantable Antenna for Brain-Machine Interfaces," *IEEE Access*, vol. 11, pp. 77980-77989, July 2023.
- [3] N. Abbas, I. A. Shah, Z. U. Abidin and H. Yoo, "Dual-Polarized Water-Substrate-Based MIMO Implantable Antenna for IoMT-Enabled Biotelemetric Devices," *IEEE Internet of Things J.*, doi: 10.1109/IJOT.2025.3570779
- [4] A. Iqbal, M. Al-Hasan, I. B. Mabrouk, and T. A. Denidni, "Deep implanted mimo antenna sensor for implantable medical devices," *IEEE Sens. J.*, vol. 23, no. 3, pp. 2105-2112, Feb 2023.
- [5] A. Iqbal, M. Al-Hasan, I. B. Mabrouk, and M. Nedil, "A compact implantable mimo antenna for high-data-rate biotelemetry applications," *IEEE Trans. Antennas Propag.*, vol. 70, no. 1, pp. 631-640, Jan 2022.
- [6] S. N. Shah, M. Zada, J. Nasir, S. M. A. Shah, E. Arneri, and H. Yoo, "Dual-band two-port mimo antenna for biomedical deep tissue communication: Design, characterization, and performance analysis," *IEEE Access*, vol. 11, pp. 104622-104632, 2023.
- [7] K. Liu, Z. Li, W. Cui, K. Zhang, M. Wang, C. Fan, H. Zheng, and E. Li, "Investigation of conformal mimo antenna for implantable devices based on theory of characteristic modes," *IEEE Trans. Antennas Propag.*, vol. 70, no. 12, pp. 11324-11334, Dec 2022.
- [8] IEEE Standard for Safety Levels With Respect to Human Exposure to Electric, Magnetic, and Electromagnetic Fields, 0 Hz to 300 GHz, IEEE Standard C95.1-2019, 2019.